# EXPERIMENT NO : 7

## RANDOM SAMPLING AND SAMPLING DISTRIBUTION

AIM:

To explore random sampling from a population and understand the concept of sampling distribution using Python in Jupyter Notebook.

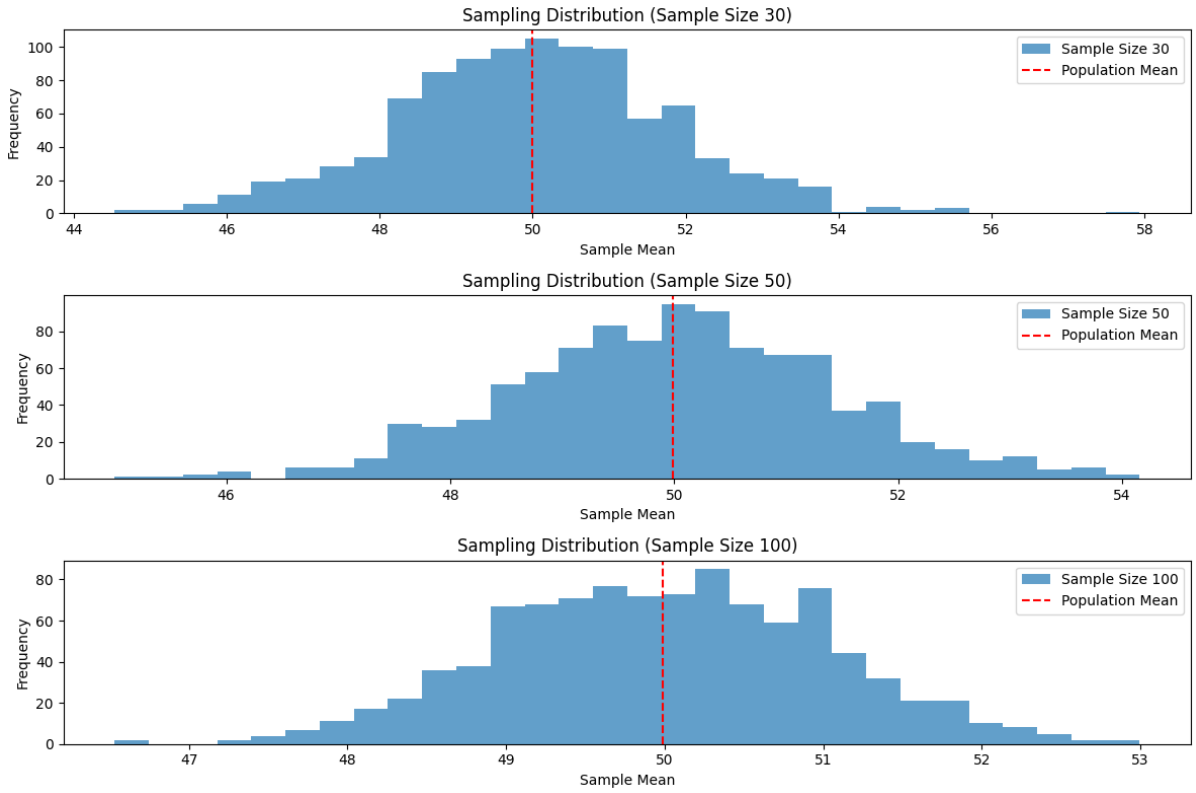
CODE:

```
import matplotlib.pyplot as plt
import numpy as np


population_mean = 50
population_std = 10
population_size = 100000
population = np.random.normal(population_mean, population_std, population_size)
sample_sizes = [30, 50, 100]
num_samples = 1000
sample_means = {}
for size in sample_sizes:
sample_means[size] = []
 for _ in range(num_samples):
 sample = np.random.choice(population, size=size, replace=False)
 sample_means[size].append(np.mean(sample))
  plt.figure(figsize=(12, 8))
 for i, size in enumerate(sample_sizes):
  plt.subplot(len(sample_sizes), 1, i + 1)
  plt.hist(sample_means[size], bins=30, alpha=0.7, label=f"Sample Size {size}")
  plt.axvline(
```

```
    np.mean(population),
    color="red",
    linestyle="dashed",
    linewidth=1.5,
    label="Population Mean",
  )
  plt.title(f"Sampling Distribution (Sample Size {size})")
  plt.xlabel("Sample Mean")
  plt.ylabel("Frequency")
  plt.legend()
  plt.tight_layout()
  plt.show()
```

OUTPUT:




RESULT :

Thus,the given data was analyzed successfully and required output was obtained.